

Supplemental Material: Geometric Origin of the Electroweak Mixing Angle

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(Dated: May 26, 2026)

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I. NORMAL-FORM THEOREM: DETAILED PROOF

We prove Theorem 1 of the main text. Let V_j be the spin- j representation of $SU(2)$, with Casimir $\sum_a T_a^{(j)} T_a^{(j)} = J \mathbf{1}$, and let $\mathcal{H}_j$ be the 2×2 block defined as Eq. (3) of the Letter, acting on $V_j \oplus \text{Im } A_j$.

Lemma 1 (Casimir factorization). *Let $\{e^a\}$ be an orthonormal basis of the cotangent space at any point of*

a parallelizable three-manifold with $SU(2)$ structure constants. Then the operator $A_j = \sum_a T_a^{(j)} \otimes e^a$ obeys $A_j^\dagger A_j = J \mathbf{1}_{V_j}$ in the orthonormal frame.

Proof. Using $\langle e^a, e^b \rangle = \delta^{ab}$,

$$A_j^\dagger A_j = \sum_{a,b} T_a^{(j)\dagger} T_b^{(j)} \delta^{ab} = \sum_a T_a^{(j)} T_a^{(j)} = J \mathbf{1}_{V_j}. \quad \square$$

Proof of Theorem 1. Axiom (i) sets the upper-left entry to zero. Axiom (ii) and Lemma 1 fix the modulus of the off-diagonal as $|\mathcal{O}_{\text{mix}}|^2 = \mu^2 J$ on the image. Choosing a phase, $\mathcal{O}_{\text{mix}} = \mu A_j^\dagger$. Axiom (iii) then gives $M_s^2 = -A_j A_j^\dagger$. Projecting to the image two-channel subspace (parameterized by $\phi \in V_j$ and the image component of a), the block becomes

$$\mu^2 \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} = \mu^2 \mathcal{A}(J), \quad (1)$$

which is unique by axiom (iv). The characteristic equation $X^2 + JX - J = 0$ has roots $X_+(J) = \frac{1}{2}(-J + \sqrt{J^2 + 4J}) > 0$ and $X_-(J) = \frac{1}{2}(-J - \sqrt{J^2 + 4J}) < 0$, with $X_+ X_- = -J$ and $X_+ + X_- = -J$. $\square$

The structural content of the four axioms is the following. Axiom (i) is gauge masslessness before symmetry breaking. Axiom (ii) is the Casimir factorization, supplied by any first-order $SU(2)$ -covariant operator on a parallelizable internal three-cycle. Axiom (iii) is the single-coupling Schur completion of Proposition 1 below. Axiom (iv) is the statement that no further dimensionless coefficient enters. These four together force $\mathcal{A}(J)$ uniquely; the weak mixing angle is then a consequence of representation theory.

II. GAUGE–HIGGS REDUCTION ON $\mathbb{CP}^1$ AND THE BARE CASIMIR

The off-diagonal $\sqrt{J}$ of $\mathcal{A}(J)$ is the matrix element of a specific physical operator built from the gauge–Higgs reduction of the six-dimensional $SU(2)$ gauge theory on $M_4 \times \mathbb{CP}^1$. We record the reduction in enough detail to fix the normalization.

A. Coset structure and line-bundle modes

The weak $\mathbb{CP}^1 \cong SU(2)/U(1)_{T_3}$ carries a homogeneous round metric of radius R . In the patch (θ, φ) the left-

invariant generators of $SU(2)$ act as Killing vectors, and the $U(1)_{T_3}$ stabilizer of the north pole is the diagonal $SU(2)$ subgroup. The minimal-magnetic-charge background on $\mathbb{CP}^1$ is the Hopf line bundle $\mathcal{O}(1)$, with first Chern number 1.

The harmonic decomposition on $\mathbb{CP}^1$ in the presence of the Hopf background is the Borel–Weil theorem:

$$H^0(\mathbb{CP}^1, \mathcal{O}(2j)) \cong V_j, \quad j = 0, \frac{1}{2}, 1, \dots \quad (2)$$

i.e. the holomorphic sections of $\mathcal{O}(2j)$ form the spin- j representation of $SU(2)$ of dimension $2j + 1$. For $j = \frac{1}{2}$ this is the Higgs doublet; for $j = 1$ it is the triplet of the same operator; $j = 0$ is the singlet.

B. The six-dimensional gauge action and its 4D content

Write the six-dimensional $SU(2)$ gauge field as $\hat{A}_M = (A_\mu(x, y), A_\alpha(x, y))$ with $\mu = 0, \dots, 3$ on M_4 and $\alpha = 1, 2$ on $\mathbb{CP}^1$. Expand the off-diagonal components $A_\alpha^\pm = (A_\alpha^1 \pm iA_\alpha^2)/\sqrt{2}$ in $\mathcal{O}(\pm 1)$ harmonics on $\mathbb{CP}^1$. The zero modes of A_α^+ form a doublet $H \in V_{1/2}$ under the surviving four-dimensional $SU(2)_W$; the zero modes of A_α^- form the conjugate doublet H^* . Together, $H = (H_+, H_0)$ is the Standard-Model Higgs.

The six-dimensional action is

$$S_6 = -\frac{1}{4g_6^2} \int d^6 X \sqrt{-g} F_{MN}^a F^{MN,a}. \quad (3)$$

Decomposing $F_{MN} = (F_{\mu\nu}, F_{\mu\alpha}, F_{\alpha\beta})$ and integrating A_α over the zero modes (2), the 4D action acquires (i) the kinetic term of the doublet H from the $F_{\mu\alpha}^2$ piece, (ii) the cross-coupling between W_μ^a and H from the cross-strength $F_{\mu\alpha} = \partial_\mu A_\alpha - \partial_\alpha A_\mu + g[A_\mu, A_\alpha]$, and (iii) a one-loop Hosotani potential for H from the $F_{\alpha\beta}^2$ piece evaluated on the KK tower.

The piece relevant for the off-diagonal of $\mathcal{A}(J)$ [Eq. (2) of the Letter] is the covariant derivative of H :

$$D_\mu H = \partial_\mu H - ig W_\mu^a T_a^{(j)} H, \quad j = \frac{1}{2}. \quad (4)$$

The generators $T_a^{(j)}$ in (4) are the *algebraic* $SU(2)$ generators acting on the line-bundle zero modes V_j , not the geometric Lie derivative on the cotangent bundle of the internal three-manifold; this is the bare-Casimir content of gauge–Higgs unification.

C. The vacuum and the gauge-boson masses

The vacuum is $\langle H \rangle = v\eta_0$ with $\eta_0 \in V_{1/2}$ the neutral component selected by hypercharge. Acting with the algebraic generators,

$$T_3^{(1/2)} \eta_0 = -\frac{1}{2} \eta_0, \quad Y \eta_0 = +\frac{1}{2} \eta_0, \quad Q \eta_0 = (T_3 + Y) \eta_0 = \frac{1}{2} \eta_0 \quad (5)$$

so the photon direction is exactly massless. The gauge-boson mass-squared matrix at the vacuum is

$$M_{ab}^2 = g^2 v^2 \langle \eta_0 | T_a^{(1/2)} T_b^{(1/2)} | \eta_0 \rangle, \quad (6)$$

whose trace is

$$\sum_a M_{aa}^2 = g^2 v^2 \langle \eta_0 | \sum_a T_a T_a | \eta_0 \rangle = g^2 v^2 J_{1/2} = \frac{3}{4} g^2 v^2, \quad (7)$$

the bare Casimir of $V_{1/2}$. Diagonalizing (6) with hypercharge mixing gives the standard $M_W^2 = g^2 v^2/4$ and $M_Z^2 = (g^2 + g'^2) v^2/4$.

D. Comparison with the standard textbook reading

In the textbook gauge–Higgs reading, the Higgs is a Wilson-line modulus, the gauge bosons get mass through the Higgs mechanism (6), and the Higgs mass comes from the one-loop Hosotani effective potential. This gives three scales: $M_W = gv/2$, $M_Z = \sqrt{g^2 + g'^2} v/2$, and $m_H = \sqrt{2\lambda} v$ with λ extracted from the loop calculation on the coset. The weak mixing angle is the embedding-coupling ratio $\sin^2 \theta_W = g'^2/(g^2 + g'^2)$, which on the $SU(3)/U(2)$ coset gives $1/4$ [5, 6]. Our setting differs: the weak coset is $SU(2)/U(1) \cong \mathbb{CP}^1$ and hypercharge comes from the $\mathbb{CP}^2$ factor with its spin^c structure, so the Manton–Fairlie $1/4$ does not apply. What does apply is the mass-squared matrix (6).

The de Vries reading uses the $(W, \delta H)$ block on the spin- j sector as a single indefinite matrix $\mathcal{A}(J)$ at one scale μ . Identifying $M_W^2 = \mu^2 X_+(3/4)$ recovers the textbook $M_W^2 = g^2 v^2/4$ through $\mu^2 = g^2 v^2/(4X_+(3/4))$. The Higgs mass is then read from the negative eigenvalue: $m_H^2 = \mu^2 |X_-(3/4)|$, with no free parameter λ . In ratio form,

$$\frac{m_H^2}{M_W^2} = \frac{|X_-(3/4)|}{X_+(3/4)} = \frac{3 + \sqrt{57}}{-3 + \sqrt{57}} \approx 2.318, \quad (8)$$

against the measured $m_H^2/M_W^2 = 2.434$. The agreement is 4.8% at the squared ratio. The same identification at $j = 1$ gives the top mass from $|X_-(2)|$; the textbook reading has no such prediction.

E. Identification with the Schur block

The identification is that the physical W and Z are the positive Schur eigenvalues of the gauge–scalar block [Eq. (7) of the Letter]. Writing the quadratic Lagrangian for $(W_\mu^a, \delta H)$ in the notation of the main Letter,

$$\mathcal{L}^{(2)} = \alpha \text{Re} \langle a, A_j \phi \rangle - \beta \langle \phi, A_j^\dagger A_j \phi \rangle, \quad (9)$$

the cross-coupling coefficient is fixed by (4) to be $\alpha = \sqrt{2} v$, and the diagonal scalar mass is the residual output of the Hosotani potential on $\mathbb{CP}^1$.

The single-coupling hypothesis (Proposition 1) is the statement that the Hosotani diagonal is $\beta = \alpha^2/g^2$, so that $c = \beta/\alpha^2 = 1$. Equivalently, the Higgs self-mass is the rank-one self-energy generated by the same operator that supplies the mixing. The structural reason this is reasonable is that gauge–Higgs unification gives the Higgs only a Hosotani-type mass, generated by the same gauge interaction (4); the open computation is the explicit one-loop coefficient on the stabilized $\mathbb{CP}^1$ with the Type IIA flux background, which sets the sign and the magnitude of β .

F. Contrast with the curl on the cotangent bundle

A natural alternative would be to identify the off-diagonal operator with the geometric Hodge curl $\star d$ on co-exact one-forms of the internal $S^3 = \text{SU}(2)$. By the Weitzenböck–Lichnerowicz identity on the unit S^3 , with Ricci tensor $\text{Ric} = 2g$,

$$\Delta_1|_{\text{co-exact}} = \nabla^* \nabla + \text{Ric} = \Delta_0 + 1 = 4J + 1 = (2j+1)^2, \quad (10)$$

so the eigenvalues of the curl on transverse one-forms are $\pm(2j+1)$. This is the curvature-shifted normalization. The algebraic second-Casimir eigenvalue of Eq. (8) of the Letter and the curl eigenvalue (10) differ by one unit of curvature: a consequence of building the off-diagonal from a curvature-bearing metric operator. The bare gauge current of gauge–Higgs unification (4) delivers the Casimir eigenvalue; an independent curvature-shifted operator would deliver $4J+1$. The numerical distinction between the two normalizations is given in Sec. IV below.

III. SINGLE-COUPLING SCHUR COMPLETION

We prove Proposition 1 of the Letter. Consider the quadratic action (9) for (a, ϕ) on the spin- j sector, with α, β real constants, and assume the order-parameter scalar ϕ satisfies the two structural hypotheses

(P1) (Flatness) The fluxless mass of ϕ vanishes: the bare quadratic term $m_0^2|\phi|^2$ is absent, so all of ϕ 's diagonal mass comes from the gauge sector.

(P2) (Single Stückelberg coupling) The only coupling between ϕ and the gauge field a is the term $\alpha a^\dagger A_j \phi + \text{h.c.}$ with the same A_j that defines the mixing.

Proposition 1 (Single-coupling Schur completion). *Under (P1) and (P2), integration of the gauge fluctuation in its own equation of motion generates the diagonal self-energy*

$$M_s^2 = -\alpha^2 A_j A_j^\dagger / g^2, \quad (11)$$

so the reduced fluctuation block on $V_j \oplus \text{Im } A_j$ is

$$\mu^2 \begin{pmatrix} 0 & A_j^\dagger \\ A_j & -A_j A_j^\dagger \end{pmatrix}, \quad \mu^2 = \frac{\alpha^2}{g^2}, \quad (12)$$

with $c = \beta/\alpha^2 = 1$.

Proof. By (P1), the bare quadratic action for ϕ has no diagonal mass. By (P2), the only quadratic term linking ϕ to a is the coupling $\alpha a^\dagger A_j \phi + \text{h.c.}$. The gauge equation of motion is therefore

$$g^2 a = \alpha A_j^\dagger \phi. \quad (13)$$

Substituting (13) back into the action gives the ϕ -only quadratic term

$$-\frac{\alpha^2}{g^2} \langle \phi, A_j A_j^\dagger \phi \rangle, \quad (14)$$

which is the rank-one self-energy (11). Hence $\beta = \alpha^2/g^2$ and $c = 1$. Projecting to the image two-channel subspace and using the Casimir factorization of Lemma 1 reduces (12) to $\mu^2 \mathcal{A}(J)$. $\square$

The structural content of (P1)–(P2) is exactly the defining feature of gauge–Higgs unification: (P1) is the Hosotani flatness of the Wilson-line modulus, and (P2) is the single $F_{\mu a}$ cross-coupling of (4). What remains open is the value of the coefficient β from the explicit one-loop Hosotani potential on the stabilized $\mathbb{CP}^1$. The proposition says that any single-coupling flat modulus has $\beta = \alpha^2/g^2$, regardless of the magnitude of α ; the magnitude of the off-diagonal squared in $\mathcal{A}(J)$ is the bare $\text{SU}(2)$ Casimir J , set by Eq. (8) of the Letter.

IV. THE WEITZENBÖCK DICHOTOMY

The bare-Casimir normalization of Eq. (8) of the Letter and the curvature-shifted normalization (10) produce distinguishable weak angles. For a deformation $\mathcal{A}_c(J) = [[0, \sqrt{J}], [\sqrt{J}, -cJ]]$, the positive root is

$$X_+(J; c) = \frac{-cJ + \sqrt{c^2 J^2 + 4J}}{2}, \quad (15)$$

and the weak angle is $\sin^2 \theta_W(c) = 1 - X_+(\frac{3}{4}; c)/X_+(2; c)$. Table I records four reference points.

The local rigidity is what makes the relation falsifiable. A one-percent error in c produces a shift in $\sin^2 \theta_W$ of 1.4×10^{-3} , an order of magnitude larger than the experimental uncertainty. Future improvement of the on-shell measurement to the 10^{-5} level would test the bare Casimir against alternative near-unity values of c .

Source	c_W at $J = 3/4$	c_Z at $J = 1$	$c_1(L_Y)$ at $J = 1$
Single-coupling Schur (algebraic)	1	1	1
Weitzenböck-shifted curl	4/3	9/8	0.22310
Conformal weight ($L_0/(k+2)$)	1/2	1/2	0.267
Standard gauge–Higgs (Manton/Fairlie)	—	—	0.36

TABLE I. The weak angle predicted by four distinct normalizations of the gauge–scalar mass operator. Only the bare SU(2) Casimir of gauge–Higgs unification reproduces the measured value. The local sensitivity at the single-coupling point $c = 1$ is $d \sin^2 \theta_W / dc = -0.140$.

V. REALIZATION IN TYPE IIA ON $M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$

The operator $\mathcal{A}(J)$ enters the linearized fluctuation spectrum of a ten-dimensional Type IIA flux compactification on $M_4 \times K_6$ with the minimal internal manifold

$$K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1. \quad (16)$$

This is the product of the smallest SU(3) coset and the smallest SU(2) coset, saturating the internal dimension six. The $\mathbb{CP}^1$ factor supplies the Borel–Weil sectors (2). The $\mathbb{CP}^2$ factor carries color and hypercharge.

A. The eleven-dimensional parent as a controlled limit

The Freund–Rubin family M^{pqr} of homogeneous spaces [4] admits the circle fibration $S^1_Z \rightarrow M^{pqr} \rightarrow \mathbb{CP}^2 \times \mathbb{CP}^1$, so K_6 is the ten-dimensional reduction of M^{pqr} . The eleven-dimensional limit is well controlled in this sense, and it supplies the linearization that allows the gauge–scalar block to be read. The integer labels (p, q, r, n) play no role in the present construction: the Bailin–Love coupling ratio is one-to-one with a four-integer family that is dense around any target, so the precision of any integer match expresses the density of the family. This overdetermination problem is the reason the present route relies on the algebraic operator $\mathcal{A}(J)$. The explicit scan demonstrating the density of integer matches is recorded in Sec. VIII.

B. The nine-dimensional electroweak endpoint

After the single-coupling scalar of Proposition 1 develops its vacuum, the weak SU(2) is broken to the electromagnetic stabilizer. Let P_{T_3} and P_Y denote the unit frame bundles of the weak weight line and the hypercharge line. The neutral character is

$$\chi_N(u_T, u_Y) = u_T^{-1} u_Y, \quad (17)$$

and the diagonal frame circle is

$$S^1_Q = \ker \chi_N \subset P_{T_3} \times P_Y, \quad (18)$$

the line bundle whose first Chern class is $c_1(L_Y) - c_1(L_{T_3})$. The infrared geometry is $M_4 \times \mathbb{CP}^2 \times S^1_Q$, and the surviving photon is the zero mode of the diagonal U(1)_Q generator $Q = T_3 + Y$. The photon is exactly massless by (5).

The global form of the surviving gauge group is $(\text{SU}(3)_c \times \text{U}(1)_Q)/\mathbb{Z}_3$. Writing the integer hypercharge $y = 6Y$, the diagonal $\mathbb{Z}_6$ generator of the Standard-Model gauge group is $\zeta_6 = (\omega_3, -1, e^{i\pi/3})$ with $\omega_3 = e^{2\pi i/3}$; on a surviving state with color triality t , weak weight $T_3 = m$, and charge $Q = m + Y$, it acts as $\omega_3^t e^{2\pi i Q}$, which reduces to $\zeta_3 = (\omega_3, e^{2\pi i/3})$ on the residual group. The descent $\mathbb{Z}_6 \rightarrow \mathbb{Z}_3$ is therefore the kinematic content of the single-coupling symmetry breaking.

C. Chirality and the role of $D = 10$

Chiral fermions are obstructed for any smooth eleven-dimensional Kaluza–Klein reduction [2]; the obstruction is removed in ten dimensions by D-branes wrapping the weak $\mathbb{CP}^1$ and carrying the Standard-Model chirality lattice [3]. The nine-dimensional infrared residue $\text{SU}(3)_c \times \text{U}(1)_Q$ is vector-like and admits no chiral obstruction. The conclusion is that the physical rung is $D = 10$: the operator $\mathcal{A}(J)$, the chiral fermions, and the single-coupling Higgs all live on the same ten-dimensional interface, with the eleven-dimensional parent supplying linearization and the nine-dimensional residue supplying the vacuum.

D. The finite Higgs sheet

The single-coupling scalar of Proposition 1 admits an almost-commutative interpretation in the sense of Connes [7, 8]. Writing the finite-direction Dirac operator $D_F = \begin{pmatrix} 0 & \Phi \\ \Phi^\dagger & 0 \end{pmatrix}$, the spectral distance to the vacuum is $d_{\text{sheet}} = 1/\|\Phi\|$ [9]. Identifying the finite sheet length with the vacuum expectation value through $\ell_H = \sqrt{2}/v_{\text{EW}}$ ties the electroweak scale to a finite-geometric datum: the Higgs vacuum amplitude is a length in the finite direction of the spectral triple. The negative-branch prediction $v_{\text{EW}} = \sqrt{2}\mu\sqrt{1+\sqrt{3}}$ [Eq. (15) of the Letter] then reads as a geometric identification: the finite sheet length and the vacuum amplitude are the same datum.

VI. THE NEGATIVE BRANCH AND THE ELECTROWEAK SCALE

The negative root $X_-(J) = -J - X_+(J)$ is an unavoidable consequence of the indefinite Schur block (12). For $J_{1/2} = \frac{3}{4}$ and $J_1 = 2$,

$$X_-(3/4) = \frac{1}{8}(-3 - \sqrt{57}) = -1.318\dots, \quad (19)$$

$$X_-(2) = -1 - \sqrt{3} = -2.732\dots, \quad (20)$$

both negative. Anchoring the overall scale μ to the Z pole through $M_Z^2 = \mu^2 X_+(2)$ gives

$$\mu = \frac{M_Z}{\sqrt{X_+(2)}} = \frac{91.19 \text{ GeV}}{\sqrt{\sqrt{3}-1}} = 106.6 \text{ GeV}, \quad (21)$$

and the W pole follows from $M_W = \mu\sqrt{X_+(3/4)} = 80.37 \text{ GeV}$, in agreement with $M_W = 80.377(12) \text{ GeV}$.

The negative-branch eigenvalues set companion scales,

$$\mu\sqrt{|X_-(3/4)|} = 122.4 \text{ GeV}, \quad (22)$$

$$\mu\sqrt{|X_-(2)|} = 176.2 \text{ GeV}. \quad (23)$$

The values (22)–(23) are within 3 GeV of the measured Higgs and top masses (125.25 GeV, 172.7 GeV), without further parameter fitting. Identifying the negative branch of the $j = 1$ sector with the electroweak vacuum amplitude through $v_{\text{EW}} = \sqrt{2}\mu\sqrt{|X_-(2)|}$ gives

$$v_{\text{EW}} = \sqrt{2}\mu\sqrt{1+\sqrt{3}} = 249 \text{ GeV}, \quad (24)$$

to be compared with the measured 246.2 GeV.

The identification of $X_-(J)$ with the order-parameter spectrum is the geometric content of the radion/Higgs reading. It is consistent within the model and agrees with data at the few-GeV level; a first-principles derivation of the negative-branch identification requires the explicit Hosotani potential beyond the quadratic order displayed in Proposition 1.

A. Orbital-velocity form and two-description correspondence

The eigenvalue equation $X^2 + JX - J = 0$ admits a kinematic interpretation. Setting $X_+ = \beta^2$ in the orbital-velocity reading,

$$\frac{\beta^2}{\sqrt{1-\beta^2}} = \sqrt{J}, \quad (25)$$

which on squaring reproduces $X^2 + JX - J = 0$. The gauge-boson masses are therefore $M_V = \mu\beta_V$ with $\beta_W = 0.7541$, $\beta_Z = 0.8556$. The bound $X_+(J) < 1$ that keeps the electroweak masses below the compactification scale is the relativistic bound $\beta < 1$.

The same eigenvalue equation $X^2 + JX - J = 0$ thus arises from two unrelated structures: the field-theoretic spin- j fluctuation block of the Kaluza–Klein reduction, and the kinematic quantization (25) of a single relativistic mode with internal orbital momentum $\sqrt{J}$. Whether the agreement reflects a duality between the two formulations, or coincidence of the underlying algebra, is open; a precise field-theory derivation of (25) from the same Kaluza–Klein action would settle it.

VII. THE 96.5 GEV PREDICTION AND THE TWO-SPIN TOWER

The same operator at higher half-integer rungs predicts a sequence of neutral spin-zero states. The next is

$$M_{3/2} = \mu\sqrt{X_+(15/4)} = 96.5 \text{ GeV}. \quad (26)$$

A persistent excess in light-Higgs searches at this mass has been reported by CMS in di-photon final states [11] and by combinations of LEP and LHC data [12]. The prediction (26) is conditional on the physical-tower truncation of $\mathcal{A}(J)$, in which all j are dynamical. The alternative finite truncation at $j \leq 1$, natural in an internal geometry of WZW level two [10], would forbid the state.

The two-spin tower is the generalization to any pair of internal spins,

$$\sin^2 \theta(j_a, j_b) = 1 - \frac{X_+(J_a)}{X_+(J_b)}. \quad (27)$$

Sample values are tabulated below; the electroweak case is the first row.

(j_a, j_b)	J_a, J_b	$\sin^2 \theta(j_a, j_b)$	Identification
$(\frac{1}{2}, 1)$	$\frac{3}{4}, 2$	0.22310	electroweak
$(\frac{1}{2}, \frac{3}{2})$	$\frac{3}{4}, \frac{15}{4}$	0.30684	prediction
$(1, \frac{3}{2})$	$2, \frac{15}{4}$	0.10778	prediction
$(\frac{1}{2}, 2)$	$\frac{3}{4}, 6$	0.34852	prediction

TABLE II. The two-spin tower $\sin^2 \theta(j_a, j_b)$ for the four lowest pairings. The electroweak entry is forced; the others are predictions of the same operator for any extended sector with higher internal spins.

VIII. THE INTEGER-FIT OVERDETERMINATION, RECORDED FOR THE AVOIDANCE OF DOUBT

The Bailin–Love coupling ratio for the homogeneous space M^{pqr} gives, for $0 < \beta < \frac{1}{2}$,

$$4\beta^3 - 6\beta^2 + \left(\frac{9}{4} + \frac{q^2}{p^2}\right)\beta - \frac{1}{2}\frac{q^2}{p^2} = 0, \quad (28)$$

and

$$\frac{g_Y^2}{g_2^2} = \frac{3}{2} \frac{p^2 n^2 (3 - 4\beta)^2 (1 + \beta)}{r^2 (1 - 2\beta)}. \quad (29)$$

A modest integer scan over $p, q \leq 30$, $r \leq 300$, $n \leq 3$ gives 138 four-tuples within 10^{-3} of the target ratio 0.287169, 12 within 10^{-4} , and 2 within 10^{-5} . The achievable values are dense near the target; the six-digit agreement of the tuple (12, 14, 91, 1) expresses the density of the family.

This overdetermination is the reason the present construction relies on $\mathcal{A}(J)$ as the weak-angle source. No

integer label is used in any derivation. The labels appear only when one wishes to embed the $\mathbb{CP}^2 \times \mathbb{CP}^1$ geometry inside a Freund–Rubin parent M^{pqr} . There they are constrained by global-form data: the $\mathbb{Z}_6$ quotient of the Standard-Model gauge group, the spin^c structure on $\mathbb{CP}^2$, and the period of the hypercharge line bundle. The constraints leave the labels underdetermined at selection precision.

IX. POLE-MASS VS. RUNNING-COUPLING READING OF $\sin^2\theta_W$

The predicted $\sin^2\theta_W = 0.22310$ is the on-shell ratio $1 - M_W^2/M_Z^2$. The relevant data point is the on-shell value 0.22305(20); the $\overline{\text{MS}}$ coupling angle $\hat{s}_Z^2 \simeq 0.23122$ at $\mu = M_Z$ is a different observable. A geometric construction of *couplings* would more naturally yield an $\overline{\text{MS}}$ -like quantity; that $\mathcal{A}(J)$ lands on the on-shell mass ratio reflects that the matrix is a *mass* operator, generated by the rank-one Schur completion of a single Stückelberg coupling.

The complex pole of the gauge-boson propagator is $s_V = M_V^2 - iM_V\Gamma_V$, and two on-shell conventions circulate: the running mass $\overline{M}_V$ and the Breit–Wigner on-shell mass M_V , differing by $M_V - \overline{M}_V \simeq \Gamma_V^2/(2M_V)$. This is 34 MeV for the Z and 27 MeV for the W , translating into a few-times- 10^{-4} ambiguity in the ratio, which is the size of the residual between 0.22310 and 0.22305. The agreement is therefore within the scheme ambiguity.

X. NUMERICAL SUMMARY

Quantity	Value (this model)	Comparison
$X_+(3/4)$	0.568729	—
$X_+(2)$	$0.732051 = \sqrt{3} - 1$	—
$\sin^2\theta_W = 1 - X_+(3/4)/X_+(2)$	0.223101	0.22305(20) [1]
M_W^2/M_Z^2	0.776898	0.77687(2)
ρ_{cust}	1	1.00031(19)
$\mu = M_Z/\sqrt{X_+(2)}$	106.6 GeV	—
$M_W = \mu\sqrt{X_+(3/4)}$	80.37 GeV	80.377(12)
$\mu\sqrt{ X_-(3/4) }$	122.4 GeV	$m_H = 125.25$
$\mu\sqrt{ X_-(2) }$	176.2 GeV	$m_t = 172.7$
$M_{3/2} = \mu\sqrt{X_+(15/4)}$	96.5 GeV	prediction
$v_{\text{EW}} = \sqrt{2}\mu\sqrt{1 + \sqrt{3}}$	249 GeV	246.2
$d\sin^2\theta_W/dc$ at $c = 1$	−0.140	rigidity slope

TABLE III. Numerical content of the construction. Experimental values from [1]. The first three rows are the load-bearing predictions; rows 4–5 are immediate consequences; rows 6–10 follow from the negative-branch identification.

XI. OPEN COMPUTATIONS

Three computations sharpen the construction beyond its present state. First, the Hosotani potential on the stabilized $\mathbb{CP}^1$ with the Type IIA flux background determines the coefficient β in (9) and so the magnitude of the negative diagonal; the single-coupling Schur completion of Proposition 1 fixes the ratio $c = 1$ but not the absolute magnitude. Second, the explicit linearized IIA reduction on $M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$ with stabilized moduli verifies that the order-parameter scalar is indeed the single-coupling flat modulus assumed by (P1)–(P2). Third, the spin^c -extended fermion content on $\mathbb{CP}^2$ supplies the chiral matter of the Standard Model, including the hypercharge assignment $Y = -\frac{1}{2}$ on the right-handed singlet that closes the determinant-one lift of the gauge group.

The Hosotani-potential coefficient, the linearized IIA reduction on the stabilized internal manifold, and the spin^c -extended fermion content are computations whose completion sharpens the construction beyond its present state. The present Letter establishes the operator-level content of the electroweak mixing angle and the on-shell mass ratio, the structural reason for the bare-Casimir normalization through gauge–Higgs unification, the single-coupling Schur completion that fixes $c = 1$, the Weitzenböck rigidity of the prediction, and the 96.5 GeV scalar that follows from the same operator.

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